

Boitumelo Khumalo

Data Analytics

BrightLearn

Exercise 4: SQL Joins Exercise:

Question 1:

▶ Run selected (1000) ✓ 1 minute ago (1s) workspace default

⋮ Edit

```
114 (904,7,701,20),
115 (905,3,705,90);
116
117 --Step 5: Query the table
118 SELECT* FROM netflix.user_table.veiwng_sessions;
119
120
121 --Question 1: Show every user who has a subscription. Match users to subscriptions.
122 SELECT A.user_id,
123        user_name,
124        B.subscription_id,
125        start_date
126 FROM netflix.user_table.users AS A
127 INNER JOIN netflix.user_table.subscriptions AS B
128 ON A.user_id=B.user_id;
129
```

Add parameter

Table +

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	1 ² user_id	A ^B user_name	1 ² subscription_id	📅 start_date
1	1	Nomvula	503	2026-03-10
2	2	David	502	2026-02-01
3	3	Anele	505	2026-04-05
4	1	Nomvula	501	2026-01-15

⬇️ 4 rows | 1.286s runtime [See performance \(1\)](#) ✓ Checked for improvements

Refreshed 1m ago

Question 2:

▶ Run selected (1000) ✓ Just now (1s) workspace default

⋮ Edit

```
124 user_name,
125 B.subscription_id,
126 start_date
127 FROM netflix.user_table.users AS A
128 INNER JOIN netflix.user_table.subscriptions AS B
129 ON A.user_id=B.user_id;
130
131 --Question 2:Show every subscription with its matching plan name and monthly price.
132 SELECT A.subscription_id,
133        A.user_id,
134        plan_name,
135        monthly_price
136 FROM netflix.user_table.subscriptions AS A
137 INNER JOIN netflix.user_table.plans AS B
138 ON A.plan_id=B.plan_id;
139
```

Add parameter

Table +

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	1 ² subscription_id	1 ² user_id	A ^B plan_name	1 ² monthly_price
1	501	1	Basic	79
2	504	6	Standard	129
3	503	1	Premium	199
4	505	3	Family	249
5	502	2	Standard	129

⬇️ 5 rows | 0.562s runtime [See performance \(1\)](#) ✓ Checked for improvements

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Question 3:

▶ Run all (1000) ✓ 1 minute ago (2s) workspace default

Edit

```
135     monthly_price
136 FROM netflix.user_table.subscriptions AS A
137 INNER JOIN netflix.user_table.plans AS B
138 ON A.plan_id=B.plan_id;
139
140 ----QUESTION 3: Show every viewing session that has a matching show. Include the show title and genre
141 SELECT A.session_id,
142        A.user_id,
143        B.show_title,
144        B.genre,
145        watch_minutes
146 FROM netflix.user_table.viewing_sessions AS A
147 INNER JOIN netflix.user_table.shows AS B
148 ON A.show_id=B.show_id;
149
150
```

Add parameter

Table	+	Q	F	I	Q	E	^	X
	1 ² session_id	1 ² user_id	A ^B show_title	A ^B genre	1 ² watch_minutes			
1	904	7	Comedy Hour	Comedy	20			
2	903	1	Crime Time	Drama	60			
3	902	2	Tech Tales	Documentary	30			
4	901	1	Comedy Hour	Comedy	45			

↓ 4 rows | 1.518s runtime [See performance \(1\)](#) ✓ Checked for improvements

Refreshed 1m ago

Question 4:

▶ Run selected (1000) ✓ 1 minute ago (1s) workspace default

Edit

```
143     B.show_title,
144     B.genre,
145     watch_minutes
146 FROM netflix.user_table.viewing_sessions AS A
147 INNER JOIN netflix.user_table.shows AS B
148 ON A.show_id=B.show_id;
149
150 ----QUESTION 4: Show every viewing session with the user who watched it. Only show sessions with a matching user
151 SELECT B.user_name,
152        B.country,
153        A.session_id,
154        A.show_id,
155        watch_minutes
156 FROM netflix.user_table.viewing_sessions AS A
157 INNER JOIN netflix.user_table.users AS B
158 ON A.user_id=B.user_id;
159
```

Add parameter

Table	+	Q	F	I	Q	E	^	X
	A ^B user_name	A ^B country	1 ² session_id	1 ² show_id	1 ² watch_minutes			
1	Nomvula	Johannesburg	903	702	60			
2	David	Cape Town	902	703	30			
3	Anele	Durban	905	705	90			
4	Nomvula	Johannesburg	901	701	45			

↓ 4 rows | 0.735s runtime [See performance \(1\)](#) ✓ Checked for improvements

Refreshed 1m ago

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Question 5:

▶ Run all (1000) ✓ 4 minutes ago (17s) workspace . default

159

160

161 --QUESTION 5: Show users along with their subscriptions, the plan name, and the price. Use only users who have both a subscription and a valid plan

162 SELECT A.user_name,

163 country,

164 C.plan_name,

165 monthly_price,

166 start_date

167 FROM netflix.user_table.users AS A

168 INNER JOIN netflix.user_table.subscriptions AS B

169 ON A.user_id=B.user_id

170 INNER JOIN netflix.user_table.plans AS C

171 ON B.plan_id=C.plan_id;

172

173

Add parameter

Table +

	A _C user_name	A _C country	A _C plan_name	1 ₃ monthly_price	start_date
1	Nomvula	Johannesburg	Premium	199	2026-03-10
2	David	Cape Town	Standard	129	2026-02-01
3	Anele	Durban	Family	249	2026-04-05
4	Nomvula	Johannesburg	Basic	79	2026-01-15

↓ 4 rows

17.495s runtime

[See performance \(1\)](#)

✓ Checked for improvements

Refreshed 4m ago

Question 6:

▶ Run selected (1000) ✓ Just now (1s) workspace . default

163 INNER JOIN netflix.user_table.plans AS C

164 ON B.plan_id=C.plan_id;

165

166

167

168 ----QUESTION 6 Show every user and any subscriptions they have. Users without subscriptions must still appear

169 SELECT A.user_id,

170 user_name,

171 B.subscription_id,

172 start_date

173 FROM netflix.user_table.users AS A

174 LEFT JOIN netflix.user_table.subscriptions AS B

175 ON A.user_id=B.user_id;

176

177

178

Add parameter

Table +

	1 ₃ user_id	A _C user_name	1 ₃ subscription_id	start_date
1	1	Nomvula	503	2026-03-10
2	2	David	502	2026-02-01
3	3	Anele	505	2026-04-05
4	4	Kabelo	null	null
5	5	Lerato	null	null
6	1	Nomvula	501	2026-01-15

↓ 6 rows

0.797s runtime

[See performance \(1\)](#)

✓ Checked for improvements

Refreshed Now

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Question 7:

▶ Run selected (1000) ✓ 1 minute ago (2s) workspace default

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```
170     user_name,
171     B.subscription_id,
172     start_date
173 FROM netflix.user_table.users AS A
174 LEFT JOIN netflix.user_table.subscriptions AS B
175 ON A.user_id=B.user_id;
176
177 ----QUESTION 7 Show every plan and the subscriptions on it. Plans with no subscribers must still appear
178 SELECT A.plan_id,
179        ..... plan_name,
180        ..... subscription_id,
181        ..... user_id
182 FROM netflix.user_table.plans AS A
183 LEFT JOIN netflix.user_table.subscriptions AS B
184 ON A.plan_id=B.plan_id;
185
```

Add parameter

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	1 ² plan_id	A ^B plan_name	1 ² subscription_id	1 ² user_id
1	10	Basic	501	1
2	11	Standard	504	6
3	12	Premium	503	1
4	13	Family	505	3
5	14	Mobile	null	null
6	11	Standard	502	2

⬇️ 6 rows 1.545s runtime [See performance \(1\)](#) ✓ Checked for improvements Refreshed 1m ago

Question 8:

▶ Run selected (1000) ✓ 1 minute ago (1s) workspace default

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```
180     subscription_id,
181     user_id
182 FROM netflix.user_table.plans AS A
183 LEFT JOIN netflix.user_table.subscriptions AS B
184 ON A.plan_id=B.plan_id;
185
186 ----QUESTION 8: Show every show and any viewing sessions on it. Shows that were never watched must still appear.
187 SELECT A.show_id,
188        ..... show_title,
189        ..... B.session_id,
190        ..... watch_minutes
191 FROM netflix.user_table.shows AS A
192 LEFT JOIN netflix.user_table.viewing_sessions AS B
193 ON A.show_id=B.show_id;
194
```

Add parameter

Table +

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	1 ² show_id	A ^B show_title	1 ² session_id	1 ² watch_minutes
1	701	Comedy Hour	904	20
2	702	Crime Time	903	60
3	703	Tech Tales	902	30
4	704	Cooking Lab	null	null
5	706	Wild Earth	null	null
6	701	Comedy Hour	901	45

⬇️ 6 rows 0.760s runtime [See performance \(1\)](#) ✓ Checked for improvements Refreshed 1m ago

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Question 9:

▶ Run all (1000) ✓ 2 minutes ago (15s) workspace default

⋮ Edit [status icons] [share]

194

195

196 --QUESTION 9: Show every viewing session and the user who watched it. Sessions referencing users that do not exist must still appear (with NULL user details).

197

198 SELECT A.session_id,

199 show_id,

200 watch_minutes,

201 B.user_id,

202 user_name

203 FROM netflix.user_table.viewing_sessions AS A

204 LEFT JOIN netflix.user_table.users AS B

205 ON A.user_id=B.user_id;

206

207

Generate (Ctrl + I)

Add parameter

Table	+	Q	F	I	Q	Q	Q	Q	Q	Q
	1 ² session_id	1 ² show_id	1 ² watch_minutes	1 ² user_id	A ^B user_name					
1	901	701	45	1	Nomvula					
2	902	703	30	2	David					
3	903	702	60	1	Nomvula					
4	904	701	20	null	null					
5	905	705	90	3	Anele					

↓ 5 rows 14.518s runtime [See performance \(1\)](#) ✓ Checked for improvements Refreshed 1m ago

Question 10:

▶ Run all (1000) ✓ 1 minute ago (2s) workspace default

⋮ Edit [status icons] [share]

207

208 --QUESTION 10: Show every user, the plan they are on (if any), and the monthly price. Users without a subscription must still appear.

209

210 SELECT A.user_name,

211 country,

212 C.plan_name,

213 C.monthly_price

214 FROM netflix.user_table.users AS A

215 LEFT JOIN netflix.user_table.subscriptions AS B

216 ON A.user_id=B.user_id

217 LEFT JOIN netflix.user_table.plans AS C

218 ON B.plan_id=C.plan_id;

219

220

Generate (Ctrl + I)

Add parameter

Table	+	Q	F	I	Q	Q	Q	Q	Q	Q
	A ^B user_name	A ^B country	A ^B plan_name	1 ² monthly_price						
1	Nomvula	Johannesburg	Premium	199						
2	David	Cape Town	Standard	129						
3	Anele	Durban	Family	249						
4	Kabelo	Pretoria	null	null						
5	Lerato	Port Elizabeth	null	null						
6	Nomvula	Johannesburg	Basic	79						

↓ 6 rows 2.474s runtime [See performance \(1\)](#) ✓ Checked for improvements Refreshed 1m ago

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Question 11:

▶ Run selected (1000) ✓ 09:46 PM (1s) workspace default

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```
216 ON A.user_id=B.user_id
217 LEFT JOIN netflix.user_table.plans AS C
218 ON B.plan_id=C.plan_id;
219
220 --QUESTION 11 Show every user and every subscription, including users without subscriptions AND subscriptions
221      referencing users that do not exist.
222
223 SELECT A.user_id,
224        A.user_name,
225        subscription_id,
226        start_date
227 FROM netflix.user_table.users AS A
228 FULL JOIN netflix.user_table.subscriptions AS B
229 ON A.user_id=B.user_id;
```

Add parameter

Table +

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	1 ² user_id	A ^B user_name	1 ² subscription_id	📅 start_date
1	5	Lerato	null	null
2	1	Nomvula	503	2026-03-10
3	3	Anele	505	2026-04-05
4	2	David	502	2026-02-01
5	4	Kabelo	null	null
6	1	Nomvula	501	2026-01-15
7	null	null	504	2026-03-20

⬇ ⬇ 7 rows | 0.875s runtime [See performance \(1\)](#) ✓ Checked for improvements

Refreshed 7m ago

Question 12:

▶ Run all (1000) ✓ 09:59 PM (12s) workspace default

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```
225      start_date
226 FROM netflix.user_table.users AS A
227 FULL JOIN netflix.user_table.subscriptions AS B
228 ON A.user_id=B.user_id;
229
230 --QUESTION 12: Show every plan and every subscription, including plans without subscribers AND any subscription
231      referencing a plan that does not exist.
232
233 SELECT plans.plan_id,
234        plan_name,
235        subscription_id,
236        user_id
237 FROM netflix.user_table.subscriptions
238 FULL JOIN netflix.user_table.plans
239 ON subscriptions.plan_id=plans.plan_id;
```

Add parameter

Table +

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	1 ² plan_id	A ^B plan_name	1 ² subscription_id	1 ² user_id
1	12	Premium	503	1
2	10	Basic	501	1
3	13	Family	505	3
4	14	Mobile	null	null
5	11	Standard	504	6
6	11	Standard	502	2

⬇ ⬇ 6 rows | 11.600s runtime [See performance \(1\)](#) ✓ Checked for improvements

Refreshed 17m ago

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Question 13:

Run selected (1000) ✓ 3 minutes ago (6s) workspace default

Edit

```
234 FROM netflix.user_table.subscriptions
235 FULL JOIN netflix.user_table.plans
236 ON subscriptions.plan_id=plans.plan_id;
237
238
239 --QUESTION 13 Show every show and every viewing session, including shows that were never watched AND sessions
referencing shows that do not exist.
240 SELECT shows.show_id,
241        show_title,
242        session_id,
243        watch_minutes
244 FROM netflix.user_table.shows
245 FULL JOIN netflix.user_table.viewing_sessions
246 ON shows.show_id=viewing_sessions.show_id;
```

Add parameter

	show_id	show_title	session_id	watch_minutes
1	704	Cooking Lab	null	null
2	703	Tech Tales	902	30
3	701	Comedy Hour	904	20
4	702	Crime Time	903	60
5	706	Wild Earth	null	null
6	701	Comedy Hour	901	45
7	null	null	905	90

7 rows 6.040s runtime See performance (1) ✓ Check Review next > Refreshed 4m ago

Question 14:

Run selected (1000) ✓ Just now (1s) workspace default

Edit

```
246
247 --QUESTION 14 Show every user and every viewing session, including users with no sessions AND sessions referencing
users who do not exist.
248
249 SELECT users.user_id,
250        user_name,
251        session_id,
252        show_id,
253        watch_minutes
254 FROM netflix.user_table.users
255 FULL JOIN netflix.user_table.viewing_sessions
256 ON users.user_id=viewing_sessions.user_id;
```

Add parameter

	user_id	user_name	session_id	show_id	watch_minutes
1	1	Nomvula	901	701	45
2	1	Nomvula	903	702	60
3	3	Anele	905	705	90
4	2	David	902	703	30
5	null	null	904	701	20
6	4	Kabelo	null	null	null
7	5	Lerato	null	null	null

7 rows 0.721s runtime See performance (1) ✓ Checked for improvements Refreshed Now

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Question 15:

▶ Run selected (1000) ✓ 2 minutes ago (11s) workspace default

Edit

```
259 --QUESTION 15 Show every user, every subscription, and every plan in one query – using FULL OUTER JOIN throughout. This
260 is the hardest question – get all gaps visible at once.
261 SELECT users.user_id,
262        .....user_name,
263        .....subscription_id,
264        .....plans.plan_id,
265        .....plan_name
266 FROM netflix.user_table.users
267 FULL JOIN netflix.user_table.subscriptions
268 ON users.user_id=subscriptions.user_id
269 FULL JOIN netflix.user_table.plans
270 ON subscriptions.plan_id=plans.plan_id;
```

Add parameter

Table +

Q F B E ^ X

	¹ ₃ user_id	^A _C user_name	¹ ₃ subscription_id	¹ ₃ plan_id	^A _C plan_name
1	1	Nomvula	503	12	Premium
2	1	Nomvula	501	10	Basic
3	3	Anele	505	13	Family
4	5	Lerato	null	null	null
5	4	Kabelo	null	null	null
6	2	David	502	11	Standard
7	null	null	504	11	Standard
8	null	null	null	14	Mobile

↓

 8 rows | 11.303s runtime [See performance \(1\)](#) ✓ Checked for improvements

Refreshed 1m ago

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Bonus challenge:

Bonus question 1:

▶ Run selected (1000) ✓ 2 minutes ago (1s) workspace default

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```
168 INNER JOIN netflix.user_table.subscriptions AS B
169 ON A.user_id=B.user_id
170 INNER JOIN netflix.user_table.plans AS C
171 ON B.plan_id=C.plan_id;
172
173
174
175 --Bonus Question 1: Which users have not subscribed to any plan?
176 SELECT users.user_id,
177        .....user_name
178 FROM netflix.user_table.users
179 LEFT JOIN netflix.user_table.subscriptions
180 ON users.user_id=subscriptions.user_id
181 WHERE subscription_id IS NULL;
182
183
```

Add parameter

Table ▾ +

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	¹ ₃ user_id	^A _C user_name
1	4	Kabelo
2	5	Lerato

⬇ ▾

2 rows | 1.253s runtime [📊 See performance \(1\)](#) ✓ Checked for improvements

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Bonus question 2:

▶ Run selected (1000) ✓ Just now (1s) workspace default

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```
287
288
289
290
291
292 --Bonus question 2: Which subscriptions reference users that do not exist in the users table?
293 SELECT subscription_id,
294        .....subscriptions.user_id
295 FROM netflix.user_table.subscriptions
296 LEFT JOIN netflix.user_table.users
297 ON subscriptions.user_id=users.user_id
298 WHERE users.user_id IS NULL;
299
```

Add parameter

Table ▾ +

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	¹ ₃ subscription_id	¹ ₃ user_id
1	504	6

⬇ ▾

1 row | 1.033s runtime [📊 See performance \(1\)](#) ✓ Checked for improvements

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Bonus question 3:

```
▶ Run selected (1000) ✓ 1 minute ago (2s) workspace default
297   ON subscriptions.user_id=users.user_id
298   WHERE users.user_id IS NULL;
299
300
301
302 --Bonus question 3: Which shows have never been watched?
303 SELECT shows.show_id,
304        show_title
305 FROM netflix.user_table.shows
306 LEFT JOIN netflix.user_table.veiwing_sessions
307 ON shows.show_id=veiwing_sessions.show_id
308 WHERE session_id IS NULL;
309
310
```

Add parameter

Table				
	¹ ₃ show_id	^A ₀ show_title		
1	704	Cooking Lab		
2	706	Wild Earth		

2 rows | 1.997s runtime [See performance \(1\)](#) ✓ Checked for improvements

Refreshed 1m ago

Bonus question 4:

```
▶ Run selected (1000) ✓ 2 minutes ago (1s) workspace default
304 show_title
305 FROM netflix.user_table.shows
306 LEFT JOIN netflix.user_table.veiwing_sessions
307 ON shows.show_id=veiwing_sessions.show_id
308 WHERE session_id IS NULL;
309
310 --Bonus question 4: Which viewing sessions reference shows that do not exist?
311 SELECT veiwing_sessions.session_id,
312        veiwing_sessions.show_id
313 FROM netflix.user_table.veiwing_sessions
314 LEFT JOIN netflix.user_table.shows
315 ON veiwing_sessions.show_id=shows.show_id
316 WHERE shows.show_id IS NULL;
```

Add parameter

Table				
	¹ ₃ session_id	¹ ₃ show_id		
1	905	705		
2	905	705		

2 rows | 0.594s runtime [See performance \(1\)](#) ✓ Checked for improvements

Refreshed 2m ago

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Bonus question 5:

▶ Run selected (1000) ✓ Just now (<1s) workspace.default

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314 LEFT JOIN netflix.user_table.shows

315 ON veiwing_sessions.show_id=shows.show_id

316 WHERE shows.show_id IS NULL;

317

318 --Bonus question 5: Which plans have no subscribers?

319

320 SELECT plans.plan_id,

321 plan_name

322 FROM netflix.user_table.plans

323 LEFT JOIN netflix.user_table.subscriptions

324 ON plans.plan_id = subscriptions.plan_id

325 WHERE subscription_id IS NULL;

326

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	plan_id	plan_name
1	14	Mobile

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1 row

0.455s runtime

[See performance \(1\)](#)

✓ Checked for improvements

Refreshed Now